

Mathematics A

Unit 1 • Chapter OL-T21 Classified

Cross-session • chapter-organised candidate

17 classified items • 59 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T21 • 17 items • 59 marks

Chapter	Items	Marks	Starts
01. OL-T21 Algebraic Fractions	17	59	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Fractions

Unit 1 • 59 marks • 17 item(s)

June 2025 | 4MA1-1H Q12 | 3 marks

Linear denominators

June 2025 | 4MA1-2H Q23 | 4 marks

Divide factorisable algebraic fractions

June 2024 | 4MA1-1H Q12 | 3 marks

Linear denominators

November 2023 | 4MA1-1H Q11 | 3 marks

Linear denominators

November 2023 | 4MA1-2H Q24 | 5 marks

Equation containing algebraic fractions

November 2024 | 4MA1-2H Q23 | 4 marks

Divide factorisable algebraic fractions

June 2022 | 4MA1-1H Q11 | 3 marks

Linear denominators

June 2022 | 4MA1-2H Q19 | 3 marks

Substitute a variable relation into a fraction

January 2020 | 4MA1-2H Q24 | 4 marks

Divide factorisable algebraic fractions

January 2021 | 4MA1-1H Q12 | 3 marks

Linear denominators

January 2021 | 4MA1-2H Q14 | 3 marks

Equation containing algebraic fractions

January 2021 | 4MA1-2H Q21 | 4 marks

Multiply factorisable algebraic fractions

June 2021 | 4MA1-1H Q21 | 4 marks

Substitute a variable relation into a fraction

November 2021 | 4MA1-2H Q12 | 2 marks

Linear denominators

May-Nov 2020 | 1H Q21 | 5 marks

Substitute a variable relation into a fraction

... 2 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 09

4MA1-1H Q12

UNIT 1**3 marks**

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

.....
(Total for Question 12 is 3 marks)

WORKING SPACE

Working Unit 25

4MA1-2H Q23

UNIT 1
4 marks

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

WORKING SPACE

Question 09

4MA1-1H Q12

MODULAR UNIT 1

3 marks

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

.....
(3)

WORKING SPACE

Question 10

4MA1-1H Q11 demand

MODULAR UNIT 1

3 marks

- 11 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.

.....

WORKING SPACE

Question 31

4MA1-2H Q24 whole

MODULAR UNIT 1

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

WORKING SPACE

Question 27

4MA1-2H Q23 M01

MODULAR UNIT 1

4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.

Show clear algebraic working.

WORKING SPACE

Question 08

4MA1-1H Q11 Q11(b)

MODULAR UNIT 1

3 marks

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

.....
(3)

WORKING SPACE

Question 26

4MA1-2H Q19 Q19

MODULAR UNIT 1

3 marks

$$19 \quad a = \frac{14}{3x - 7} \quad x = \frac{7}{4y - 3}$$

Express a in the form $\frac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

 $a = \dots\dots\dots$

WORKING SPACE

24 Express

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

YOUR WORKING

12 (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

(3)

YOUR WORKING

14 (a) Solve $\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$

Show clear algebraic working.

$a =$
(3)

YOUR WORKING

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

YOUR WORKING

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

YOUR WORKING

(b) Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

(2)

YOUR WORKING

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

YOUR WORKING

13 Solve $\frac{x+3}{4} - \frac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$

(Total for Question 13 is 3 marks)

YOUR WORKING

23 $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 23 is 3 marks)

YOUR WORKING
